

Empirical Proof Record: VALIDATED

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Proof ID: 4cafed39e1c6876eb7db56a90c1a22cc...

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1. Claim

Across 100 random (k, n) binomial pairs (seed=20260518, max_n=1000), scipy's "binomtest.proportion_ci(method='wilson')" and statsmodels's "proportion_confint(method='wilson')" produce 95% CI bounds that agree to $\leq 1e-09$ on both lower and upper endpoints. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer.)

- **Operationalisation:** Generate N pairs (k, n) with $k \sim \text{Uniform}(0, n)$, $n \sim \text{Uniform}(2, \text{max_n})$. For each pair, compute the Wilson CI at the configured confidence under both backends. Compute $\text{max_abs_diff} = \max \text{ over pairs of } \max(|\text{low_scipy} - \text{low_sm}|, |\text{high_scipy} - \text{high_sm}|)$. VALIDATED iff $\text{max_abs_diff} \leq \text{tolerance}$.
- **Threshold:** $\text{max_absolute_ci_difference} \leq 1e-09 \text{ proportion}$
- **H0:** scipy and statsmodels disagree on the Wilson CI by more than the floating-point tolerance — at least one implementation has a defect.
- **H1:** Both libraries evaluate the same closed-form Wilson CI and produce identical bounds to floating-point tolerance.

2. Verdict

VALIDATED — observed 3.3306690738754696e-16

100 pairs checked; $\max |\text{low_scipy} - \text{low_sm}| = 1.110e-16$; $\max |\text{high_scipy} - \text{high_sm}| = 3.331e-16$; $\max \text{ overall} = 3.331e-16 (\leq 1e-09 \text{ tol})$

3. Evidence

Pillar	Statistic	Value	95% C
wilson_ci_scipy_vs_statsmodels	max_absolute_ci_difference	3.3306690738754696e-16	-

4. Pre-registration

- Registered at: 2026-05-24T03:31:26.141625+00:00
- Config hash: ab8fb015887c4a2ac85c8b2ffe1e676a...
- Data hash: ab8fb015887c4a2ac85c8b2ffe1e676a...

5. Signature

45e3a5a9b9b477146a5519dba1b989bf2df9d7293cb7e3404a41141513956736